



COMPUTER NETWORKS

2/19/2026

AGENDA

-  **About Us**
-  **Course Syllabus and Lab Outlines**
-  **Course Objectives**
-  **Course Overview**
- **Lecture Notes**

General Introduction

Networking Today No Boundaries



- Advancements in networking technologies are helping create a **world without boundaries**.
- The immediate nature of communications over the Internet **encourages** global communities.

NETWORK CLASSIFICATION

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- **Why?**

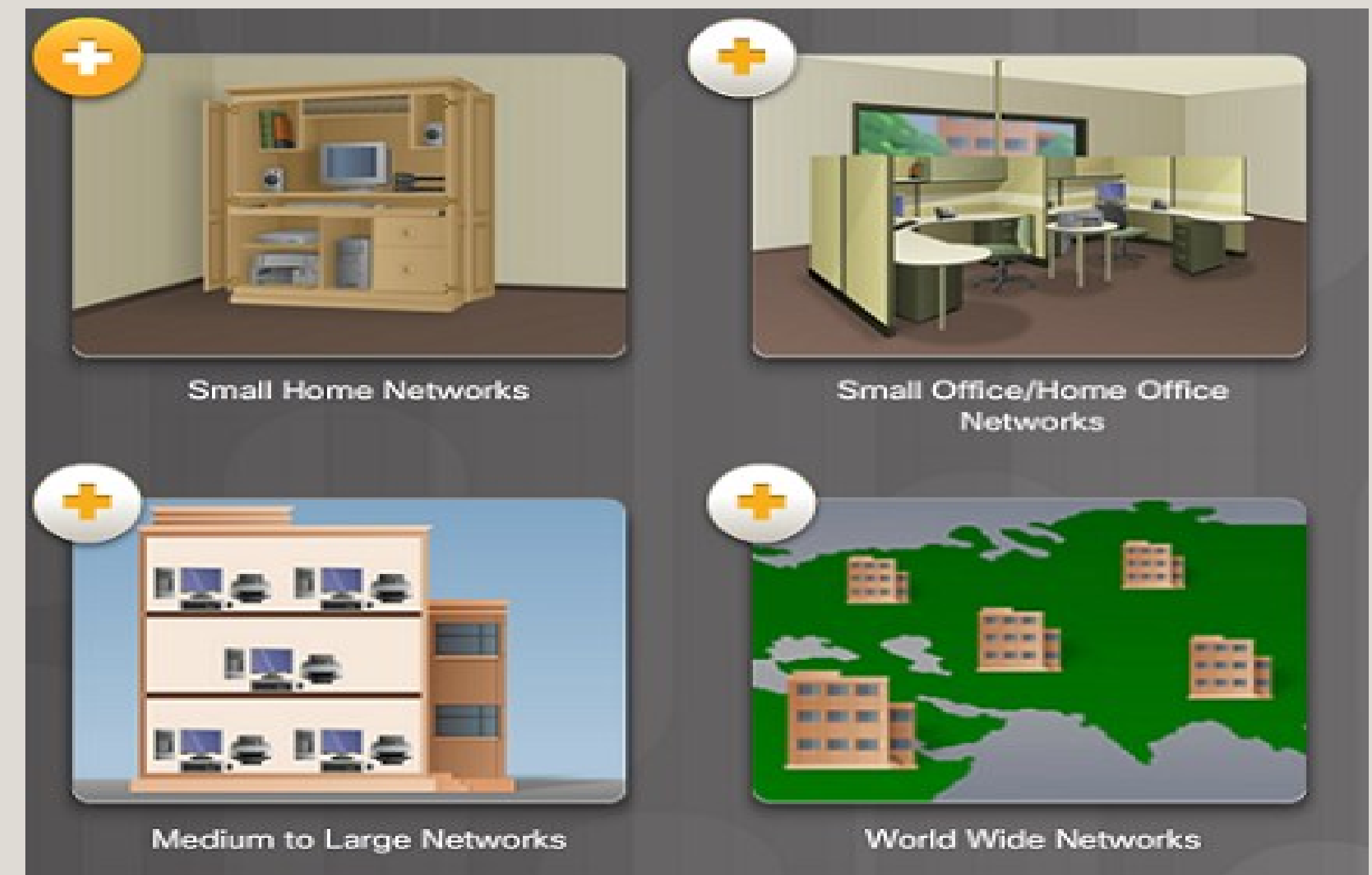
- ☐ Large Set of devices
- ☐ Devices are too far apart

- **Network Types**

- ☐ Private Area Network (PAN)
- ☐ Local Area Network (LAN)
- ☐ Metropolitan Area Network (MAN)
- ☐ Wide Area Network (WAN)
- ☐ Storage Area Network (SAN)

- **Two broad categories of communications networks**

- ☐ Wide Area Network (WAN)
- ☐ Local Area Network (LAN)
- ☐ Metropolitan Area Network

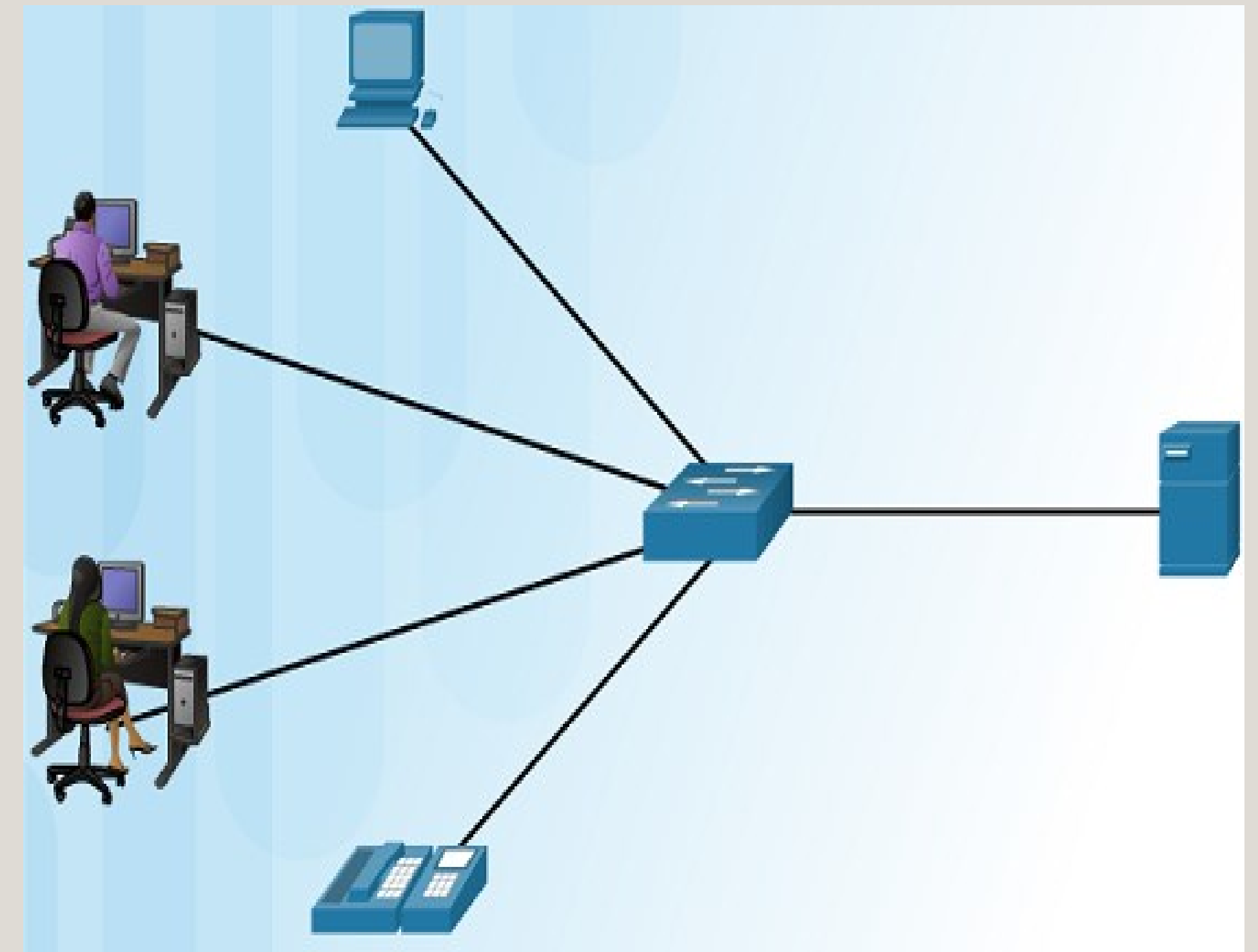


LOCAL AREA NETWORK - LAN

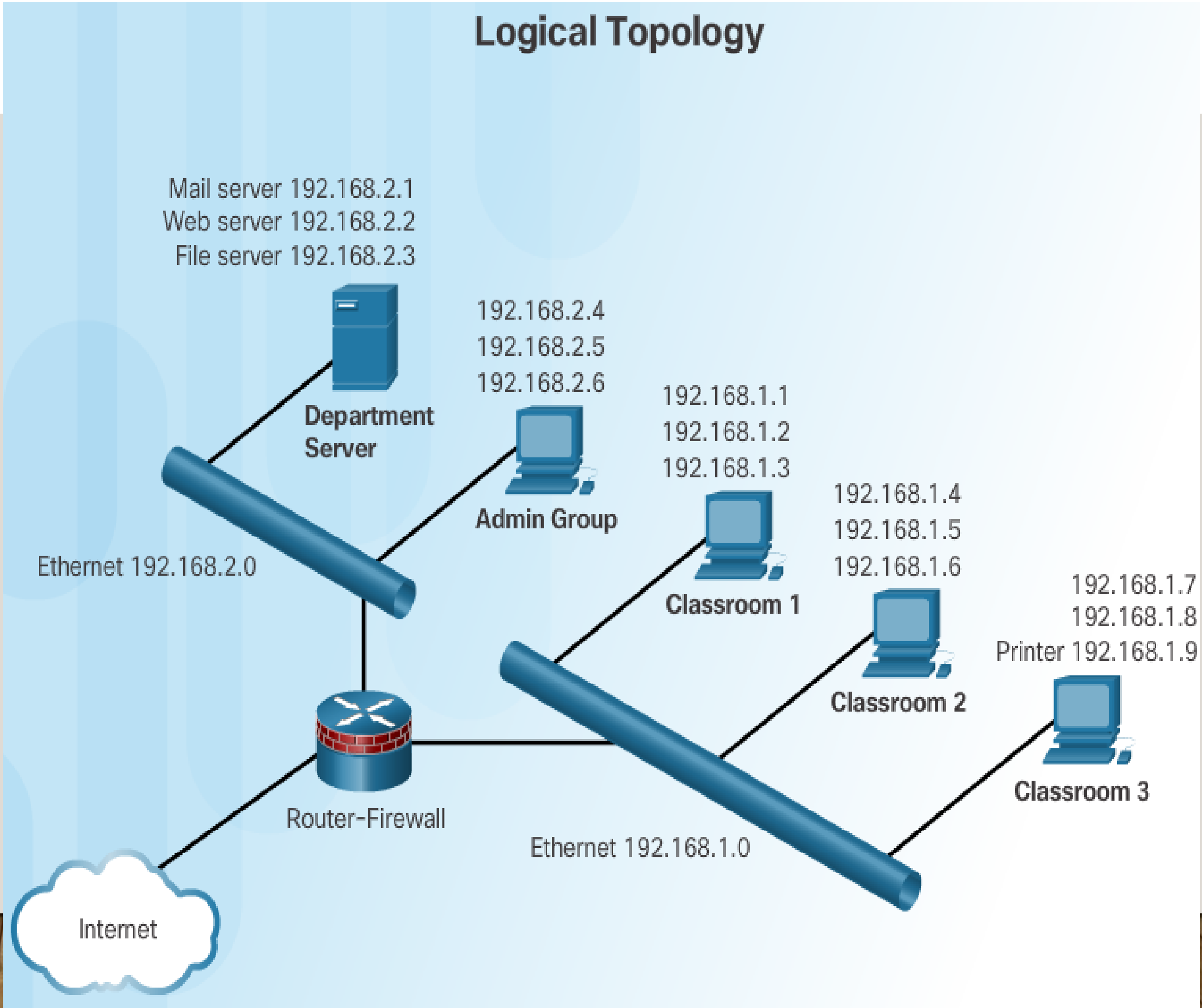
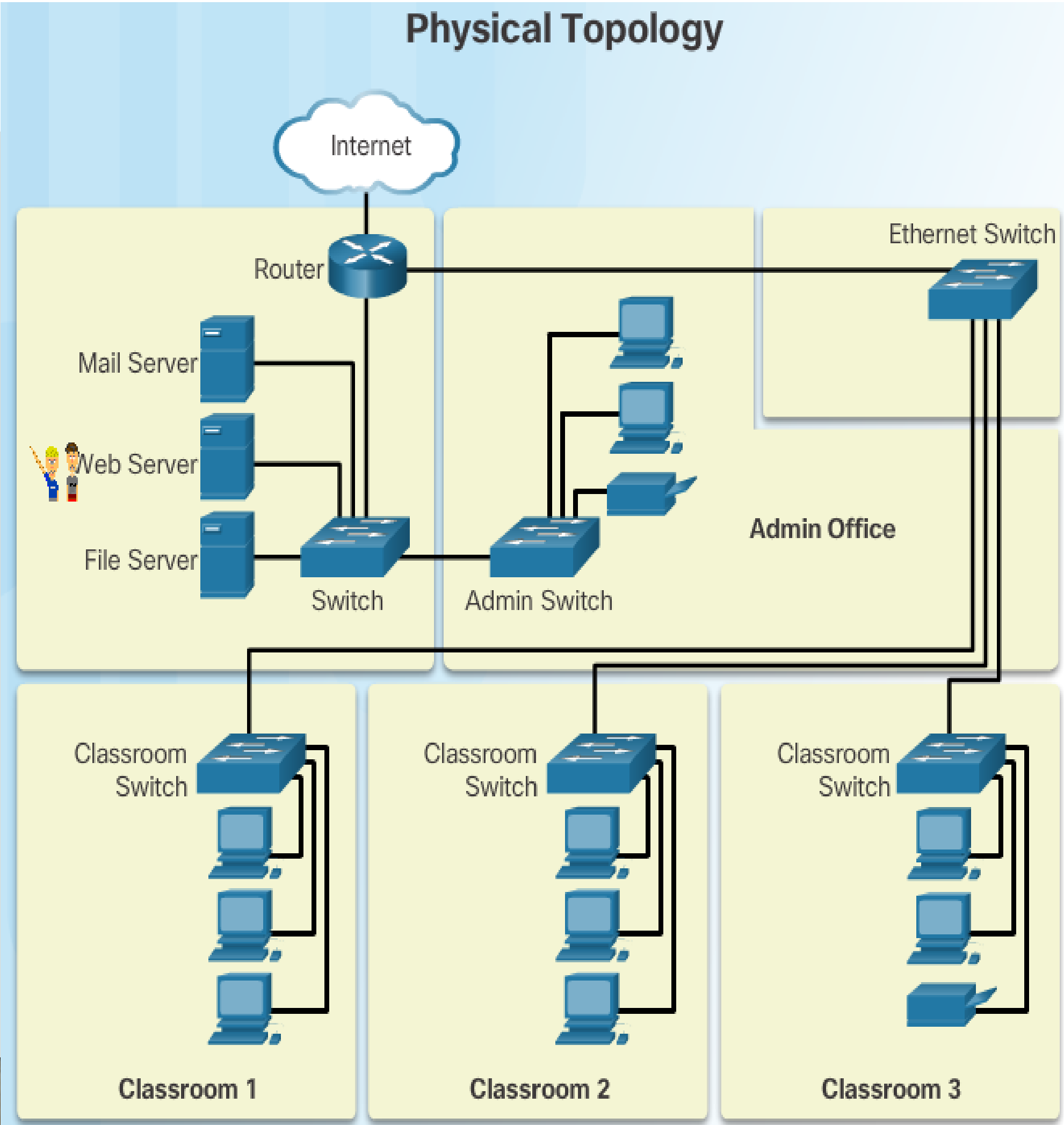
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- **Support Small geographical area**
 - ❑ Smaller Scope
 - ❑ Spans a small geographic area such as a home, school, office building, or campus.
- **Usually** owned by same organization as attached devices
- Data rates much higher (higher bandwidth)
- Usually broadcast systems
- Now some switched system and ATM are being introduced



LAN – TOPOLOGIES



Wide Area Network - WAN

- **Support Large geographical area**

- ❑ WANs interconnect LANs over wide geographical areas such as between **cities, states, or countries.**

- Crossing public rights of way
- Rely in part on common carrier circuits
- Usually administered by multiple service providers.
- **WANs typically provide slower speed links between LANs.**

- **Alternative technologies**

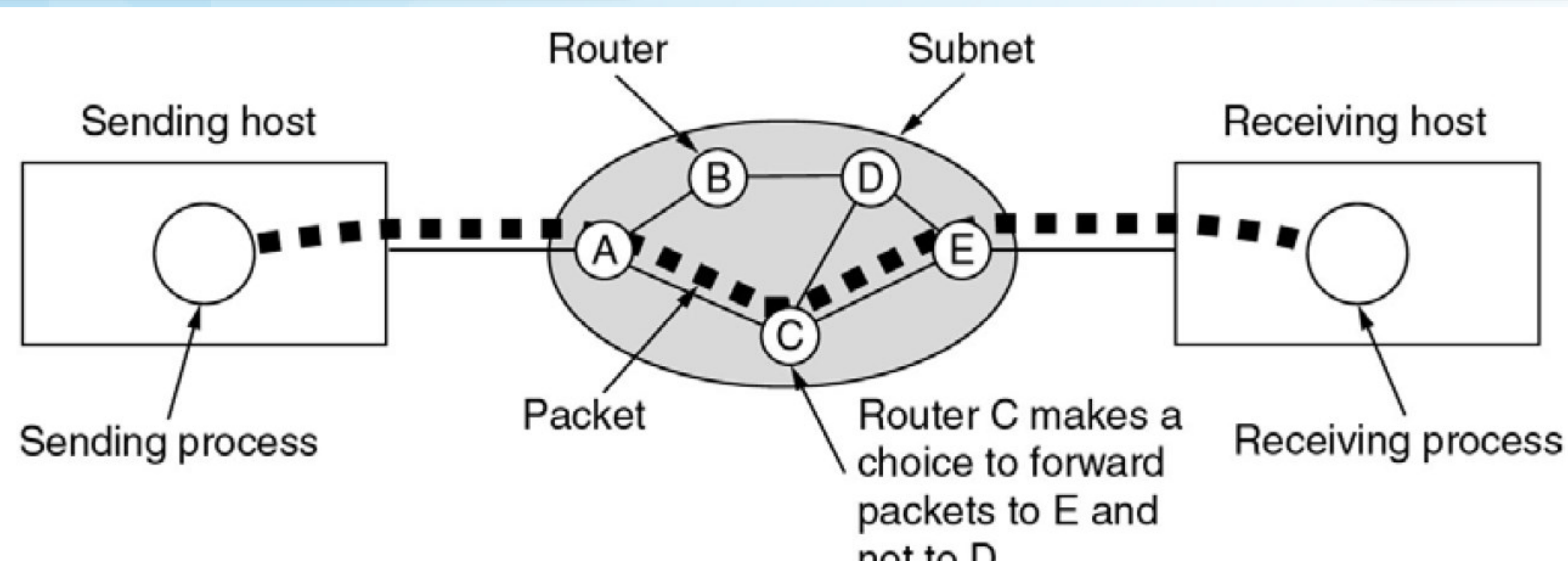
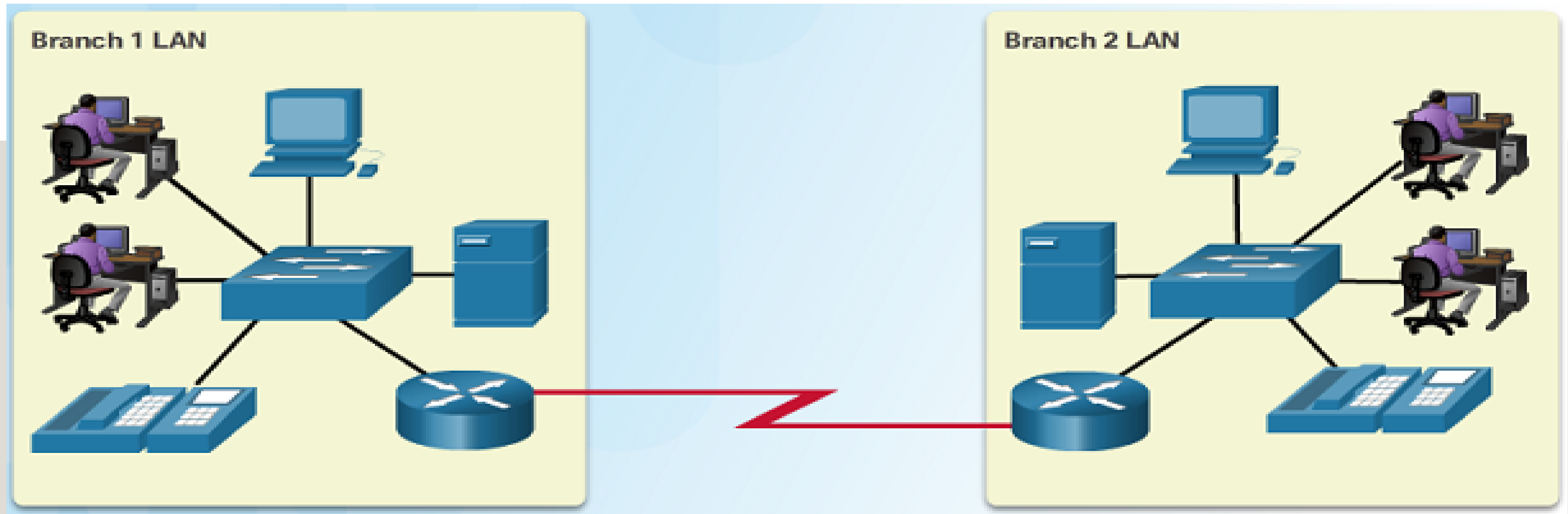
- ❑ Circuit switching
 - ❑ Packet switching
 - ❑ Frame relay
 - ❑ Asynchronous Transfer Mode (ATM)



Wide Area Network - WAN

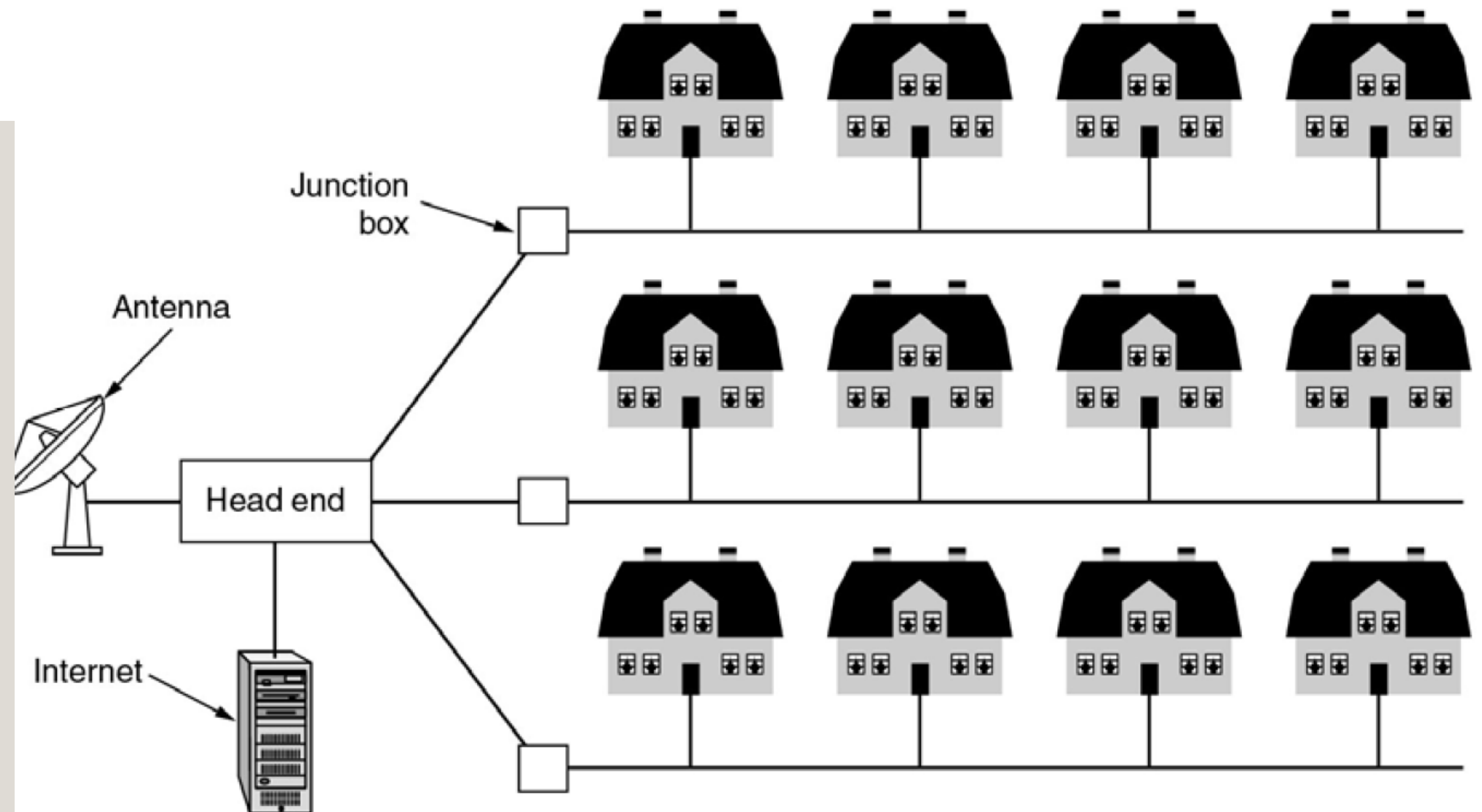
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Metropolitan Area Network

- Middle between LAN and WAN
- Private or public network
- Support High speed in large area



Classification By scale

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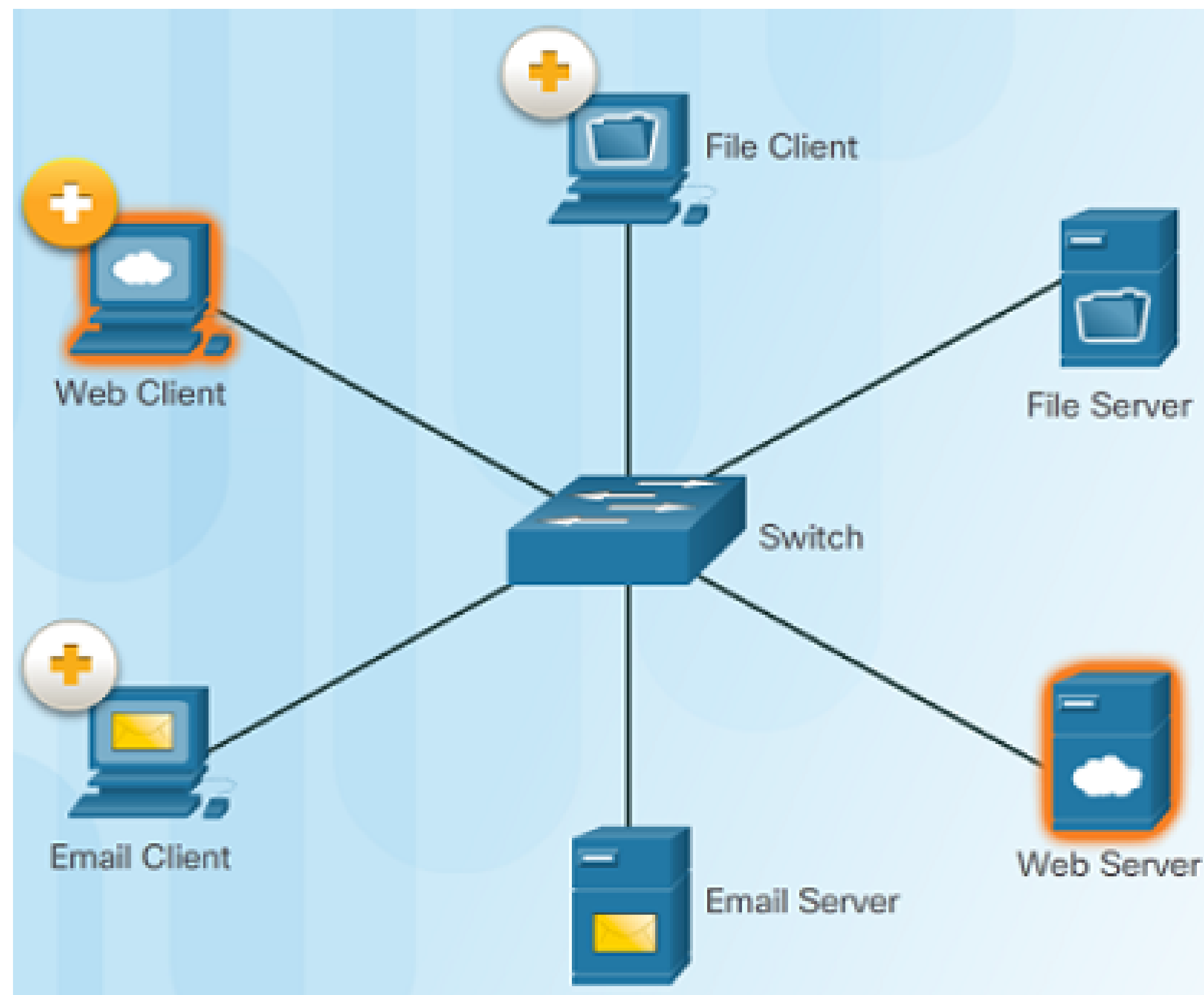
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Classification of Networks by Scale

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	
100 m	Building	
1 km	Campus	Local area network
10 km	City	
100 km	Country	Metropolitan area network
1000 km	Continent	
10,000 km	Planet	Wide area network
		The Internet

Classification By Transmission

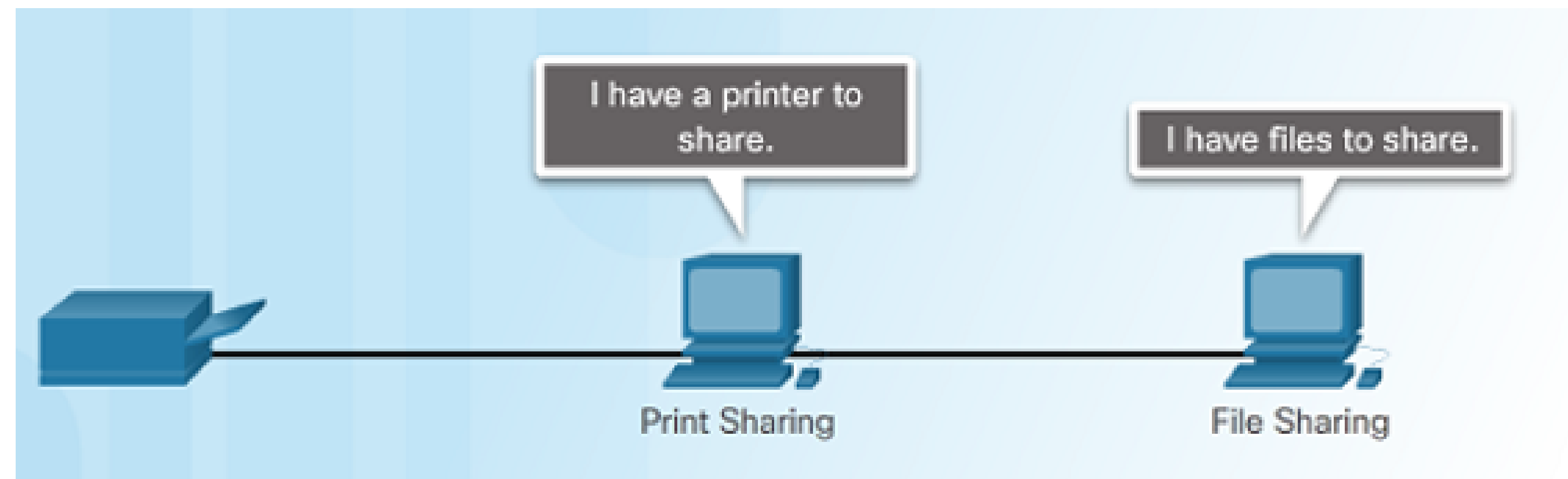
Clients and Servers



- Every computer connected to a network is called a host or end device.
- Servers are computers that provide information to end devices on the network. For example, email servers, web servers, or file server
- Clients are computers that send requests to the servers to retrieve information such as a web page from a web server or email from an email server.

Classification By Transmission

Peer-to-Peer



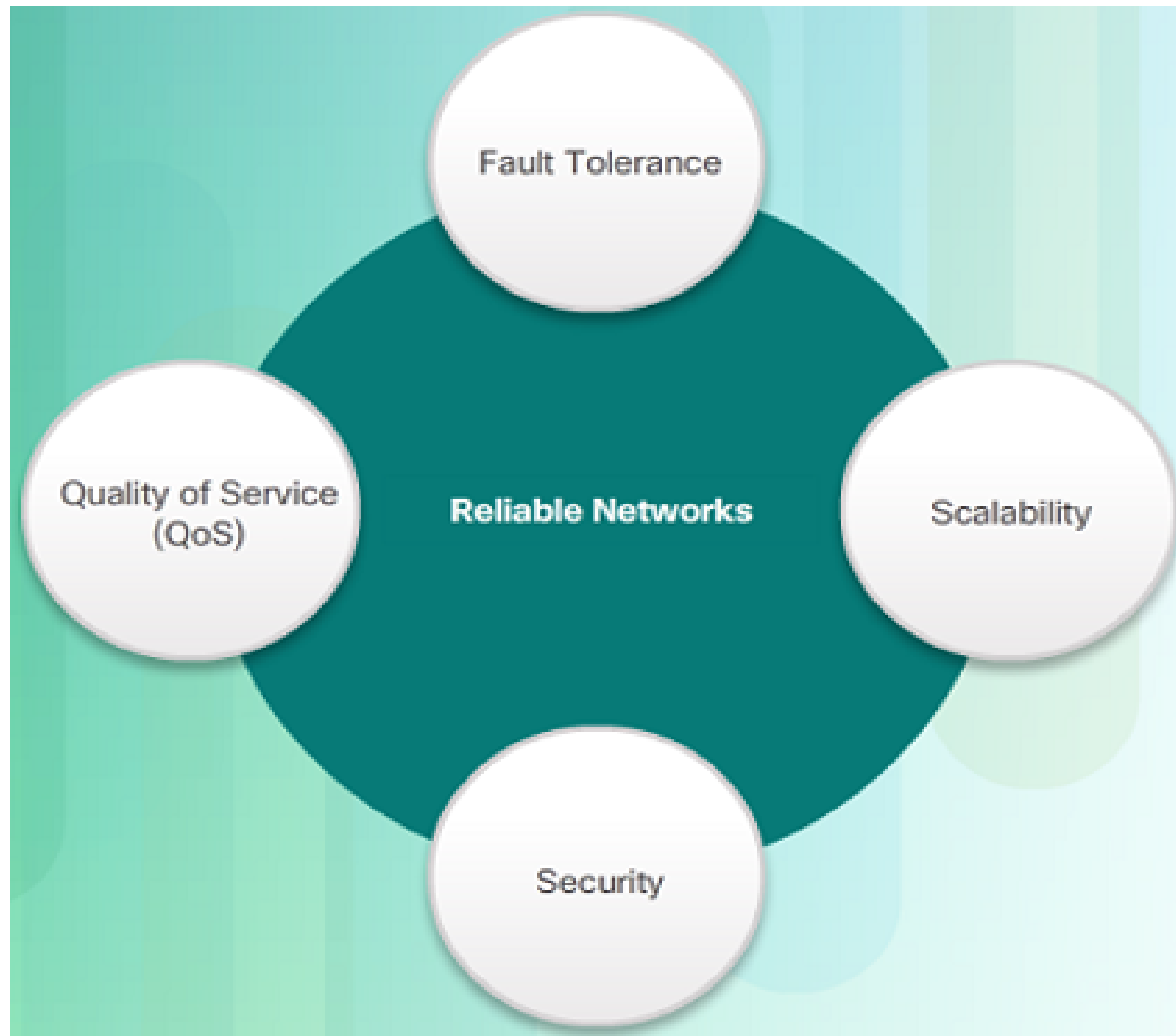
- Client and server software usually run on separate computers.
- However, in small businesses or homes, it is typical for a client to also function as the server. These networks are called peer-to-peer networks.
- Peer-to-peer networking advantages: easy to set up, less complex, and lower cost.
- Disadvantages: no centralized administration, not as secure, not scalable, and slower performance.

Network Reliable Arch.

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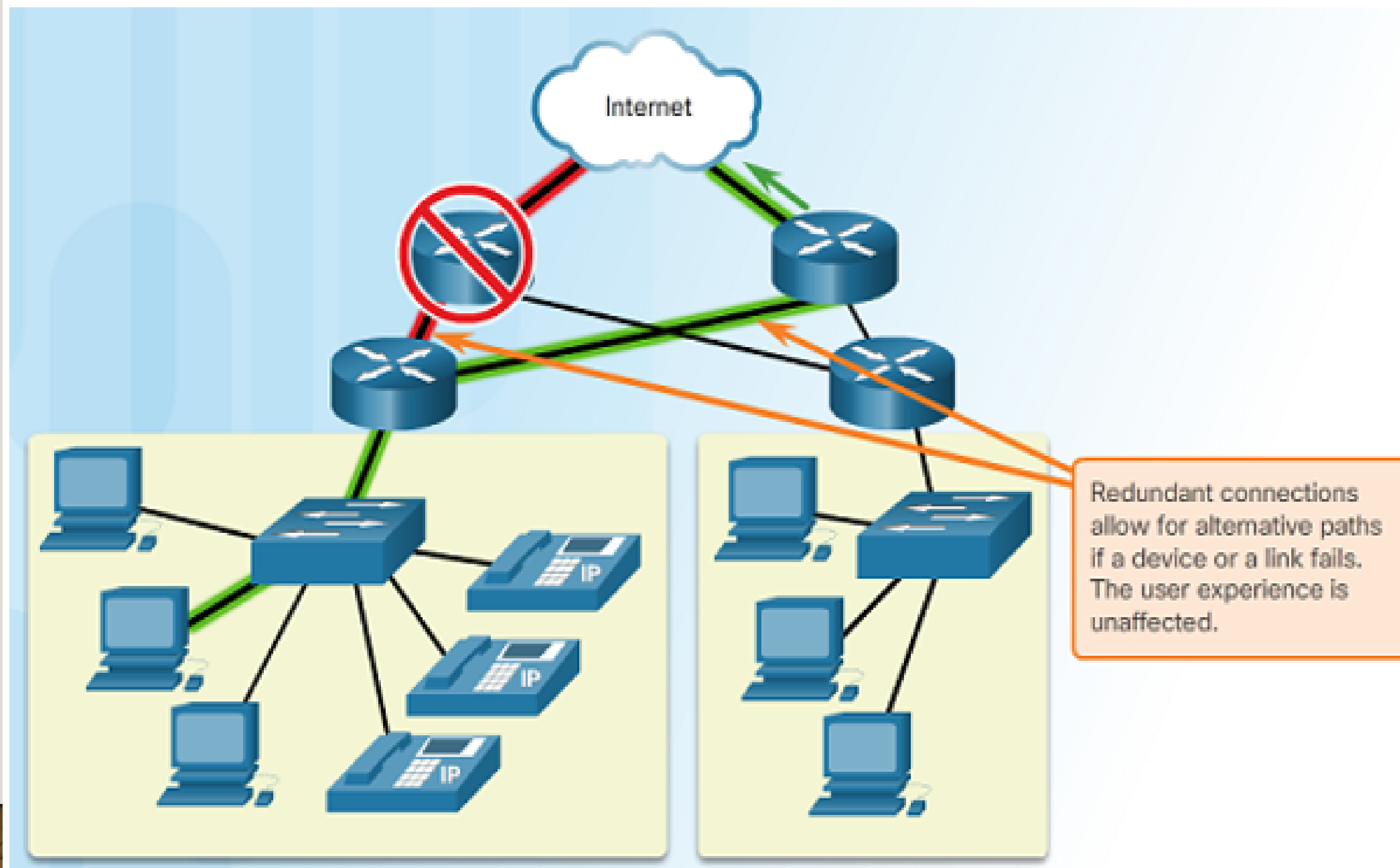
Network Architecture



- Network Architecture refers to the technologies that support the infrastructure that moves data across the network.
- There are four basic characteristics that the underlying architectures need to address to meet user expectations:
 - Fault Tolerance
 - Scalability
 - Quality of Service (QoS)
 - Security

Network Reliable Arch.

Fault Tolerance



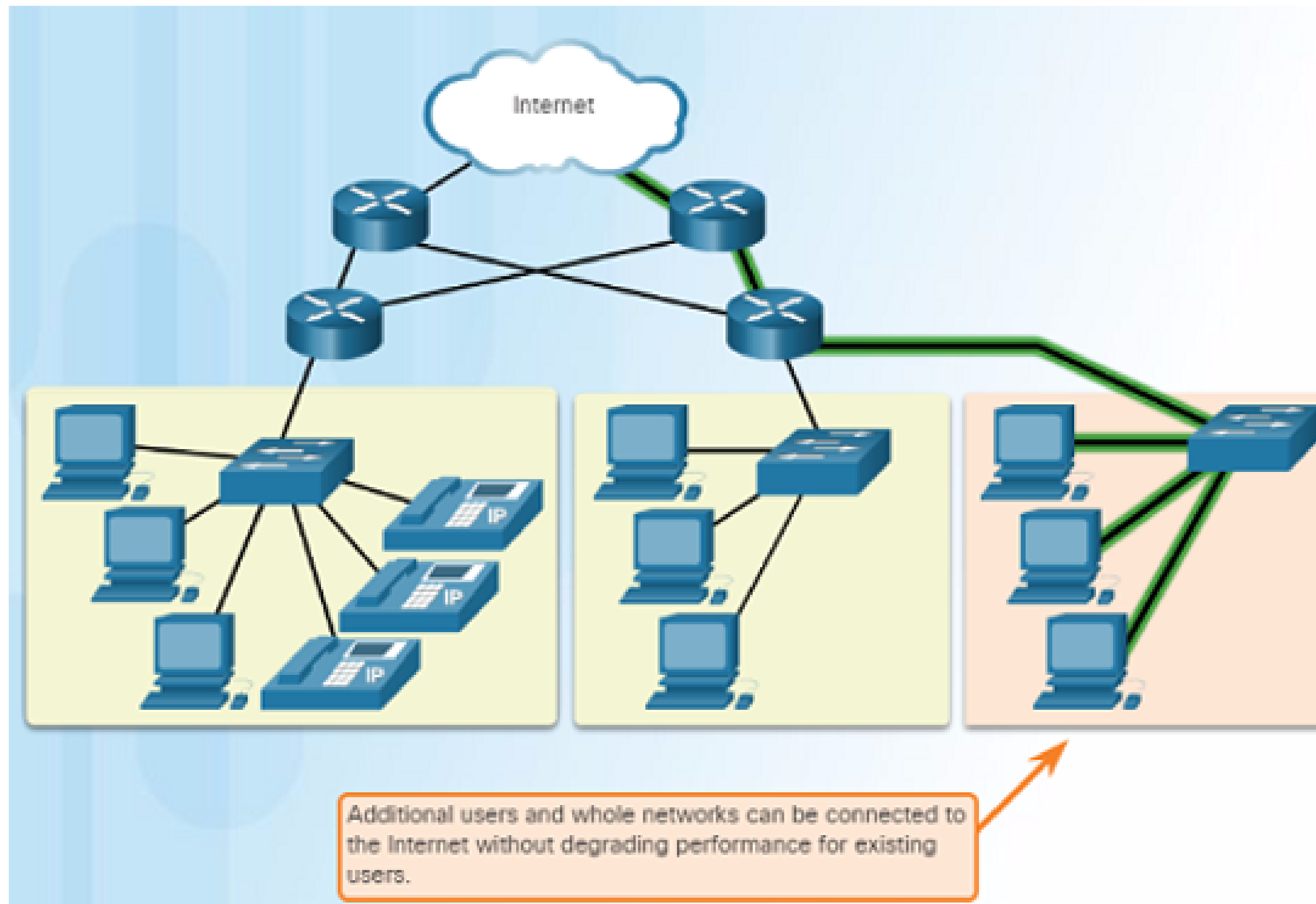
- A fault tolerant network limits the impact of a failure by limiting the number of affected devices.
- Multiple paths are required for fault tolerance.
- Reliable networks provide redundancy by implementing a packet switched network. Packet switching splits traffic into packets that are routed over a network. Each packet could theoretically take a different path to the destination.
- This is not possible with circuit-switched networks which establish dedicated circuits.

Network Reliable Arch.

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Scalability

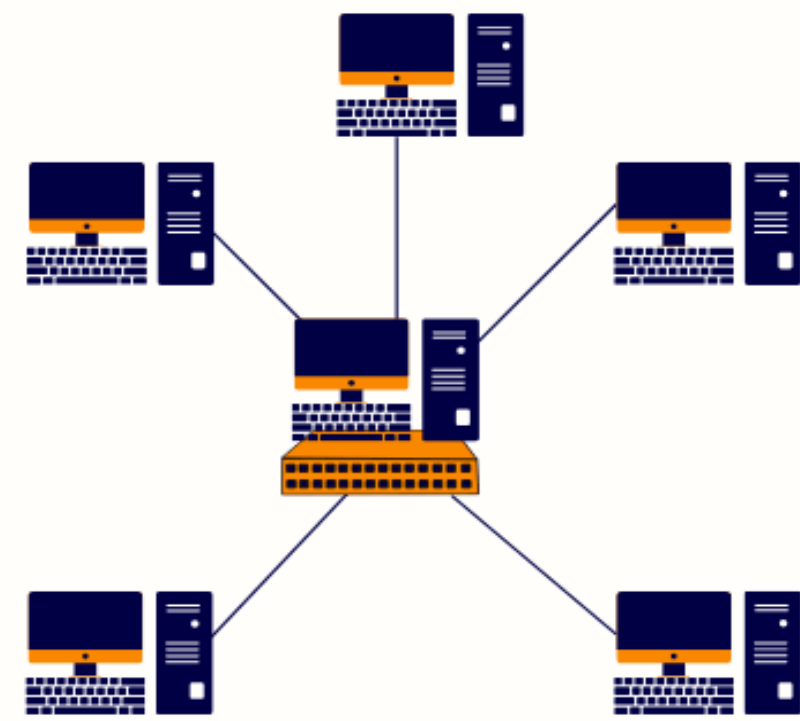


- A scalable network can expand quickly and easily to support new users and applications without impacting the performance of services to existing users.
- Network designers follow accepted standards and protocols in order to make the networks scalable.

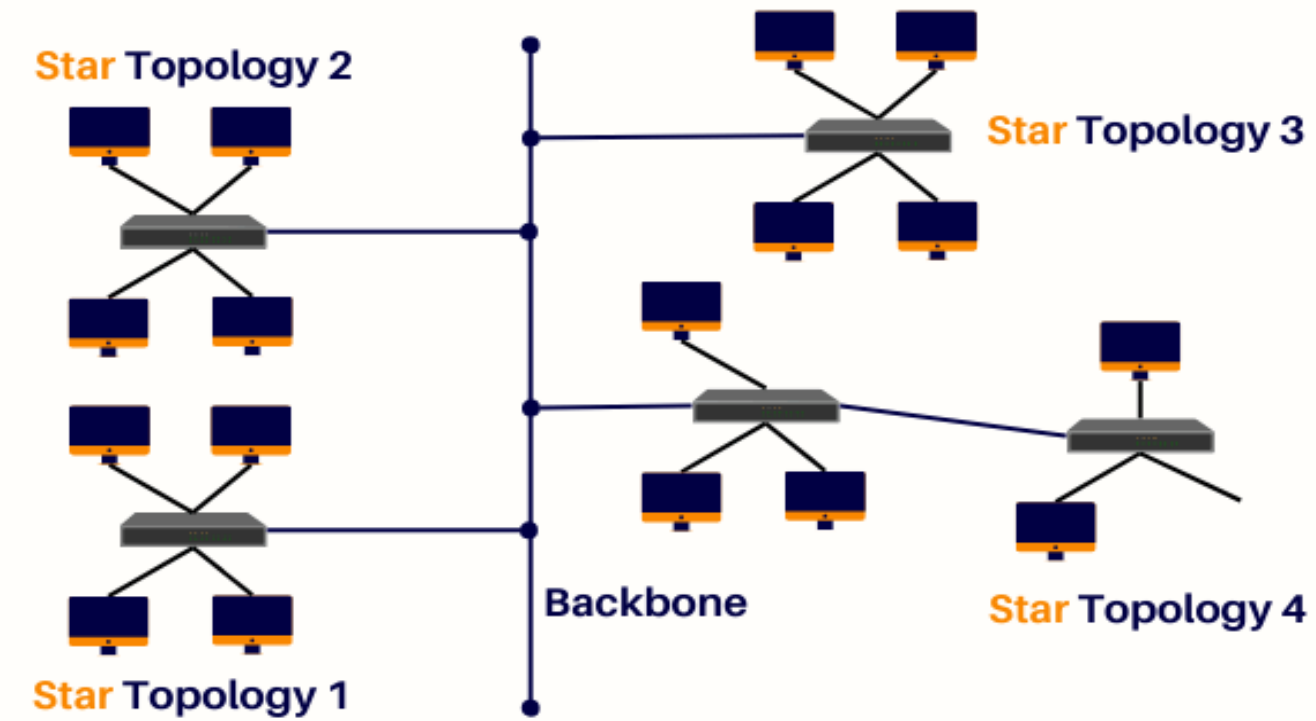
LAN – Topologies

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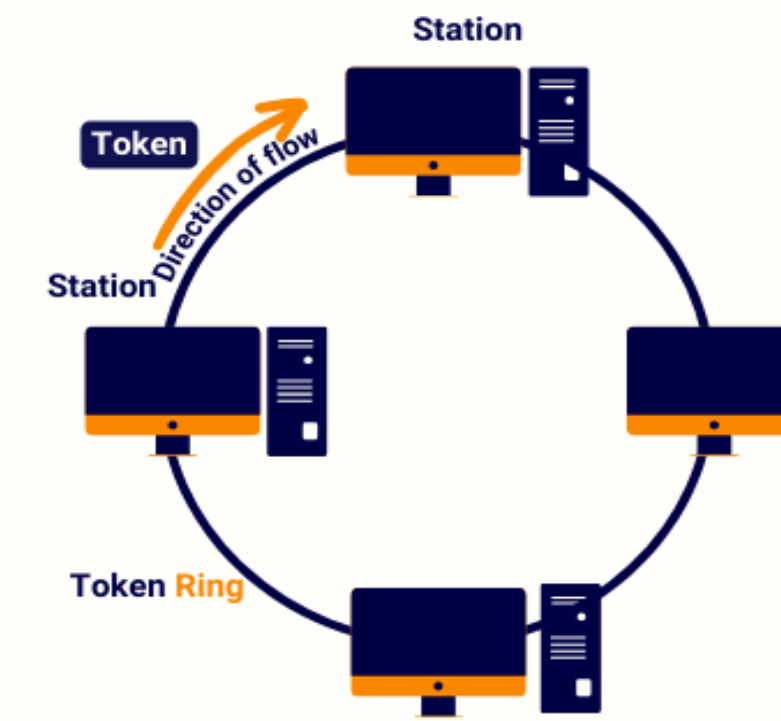
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Star



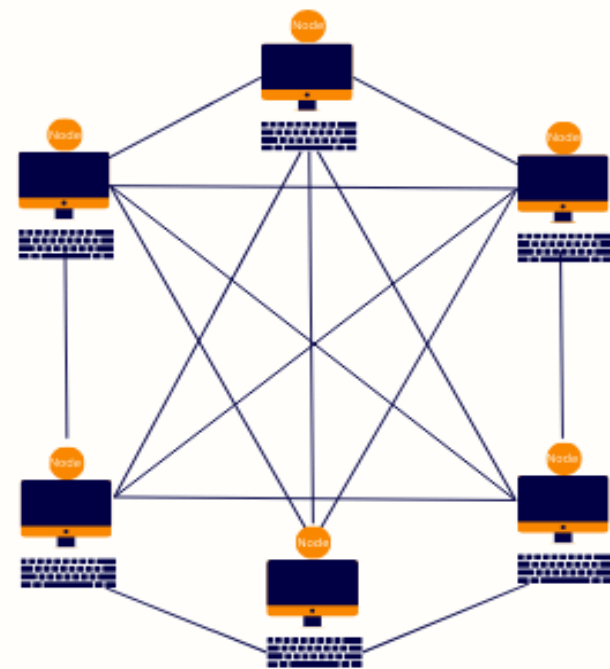
Bus



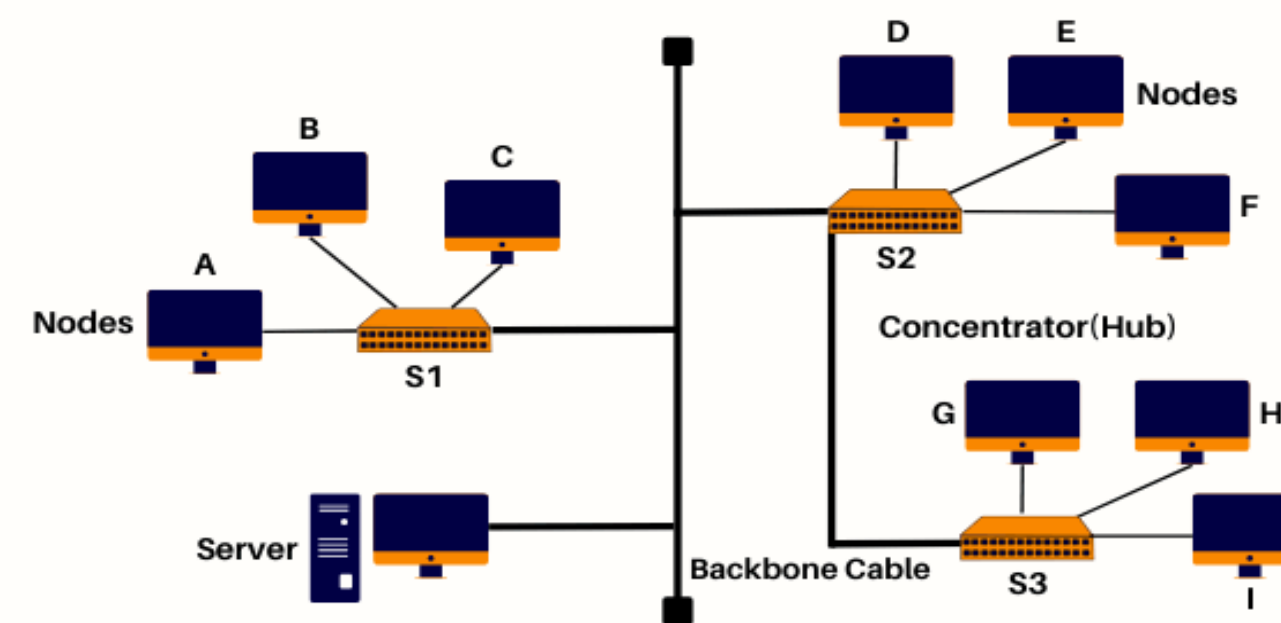
Ring

➤. Types of Network Topologies ➤.

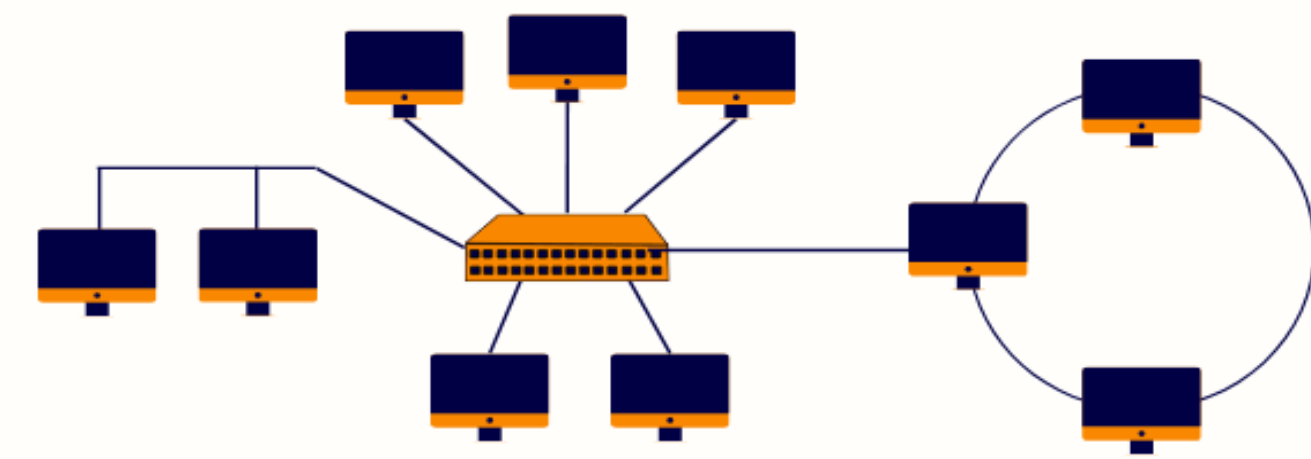
Mesh



Tree



Hybrid

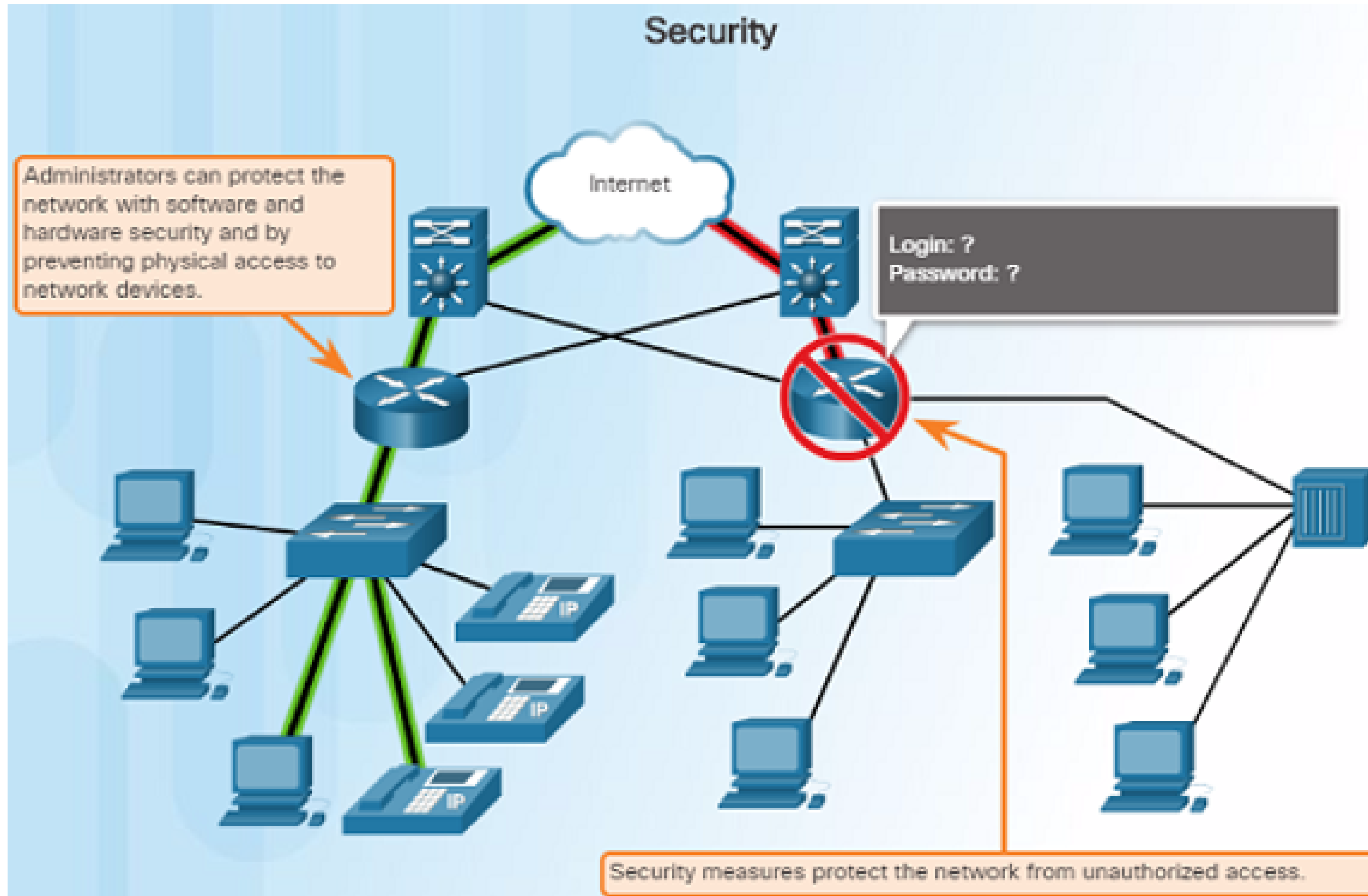


Network Reliable Arch.

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Security



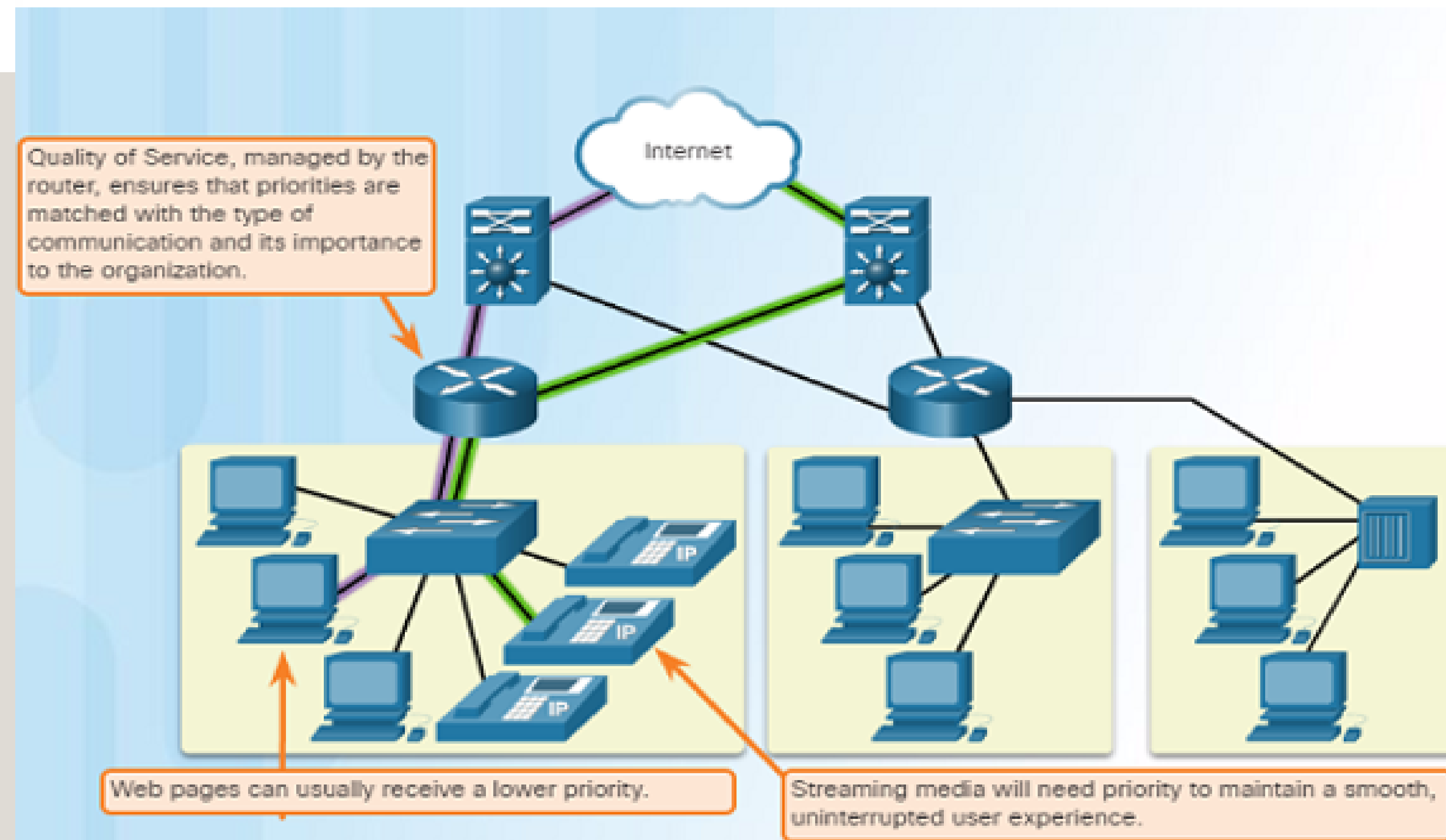
- There are two main types of network security that must be addressed:
 - Network infrastructure security
 - Physical security of network devices
 - Preventing unauthorized access to the management software on those devices
 - Information Security
 - Protection of the information or data transmitted over the network
- Three goals of network security:
 - Confidentiality – only intended recipients can read the data
 - Integrity – assurance that the data has not be altered with during transmission
 - Availability – assurance of timely and reliable access to data for authorized users

Network Reliable Arch.

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Quality of Service

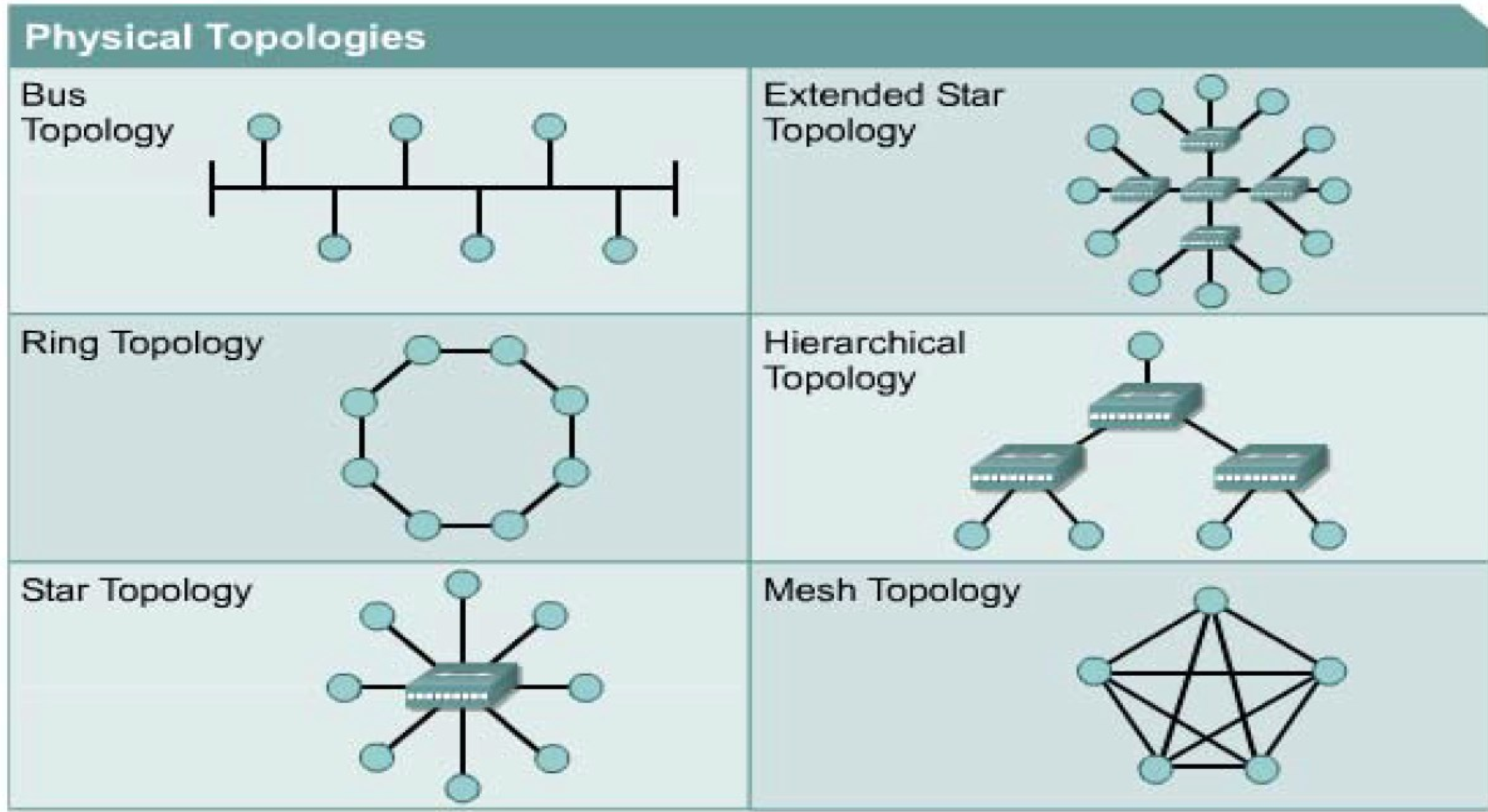


- Voice and live video transmissions require higher expectations for those services being delivered.
- Have you ever watched a live video with constant breaks and pauses? This is caused when there is a higher demand for bandwidth than available – and QoS isn't configured.
- Quality of Service (QoS) is the primary mechanism used to ensure reliable delivery of content for all users.
- With a QoS policy in place, the router can more easily manage the flow of data and voice traffic.

LAN – Topologies

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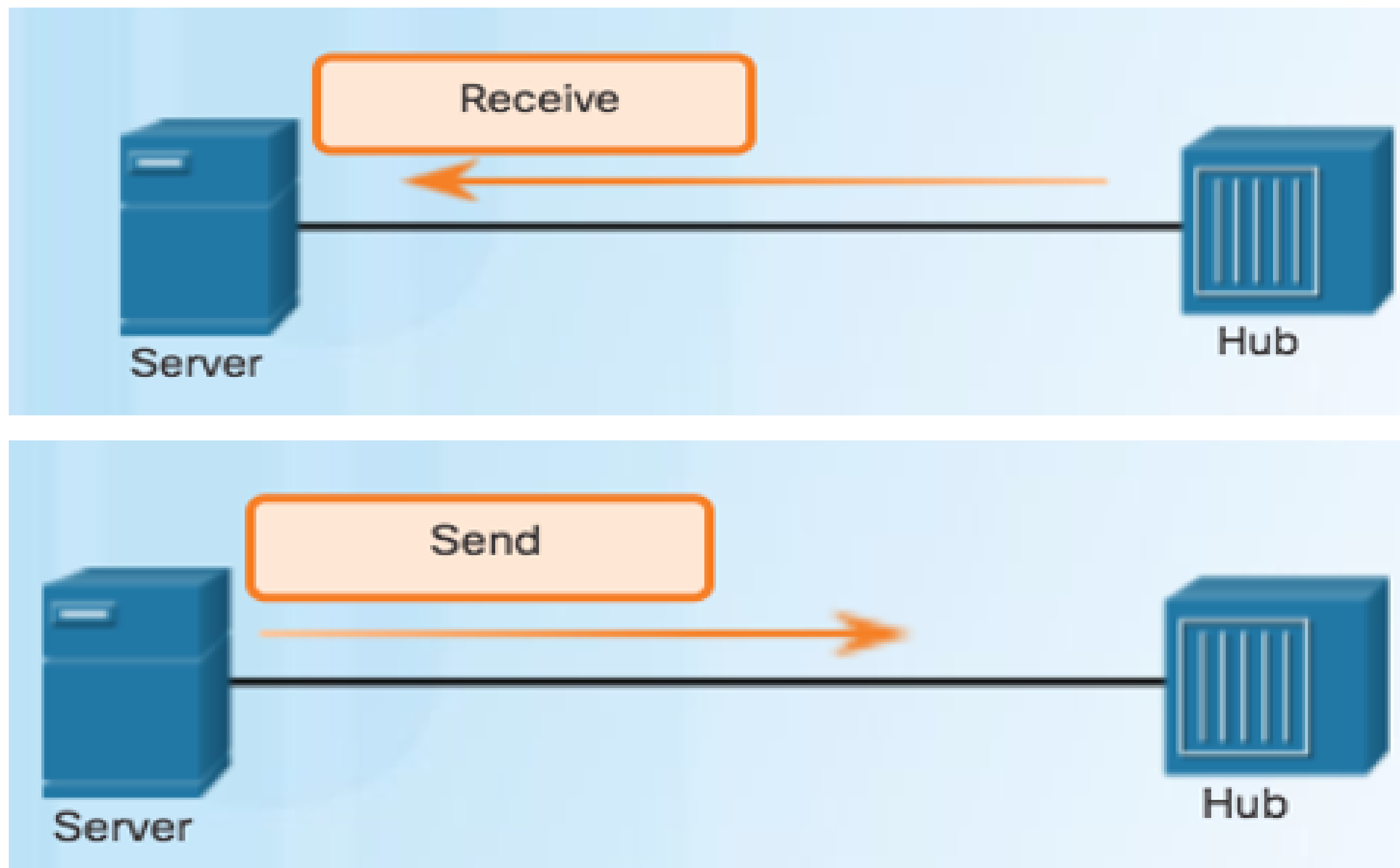


LAN – Communication

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LAN Topologies Half and Full Duplex



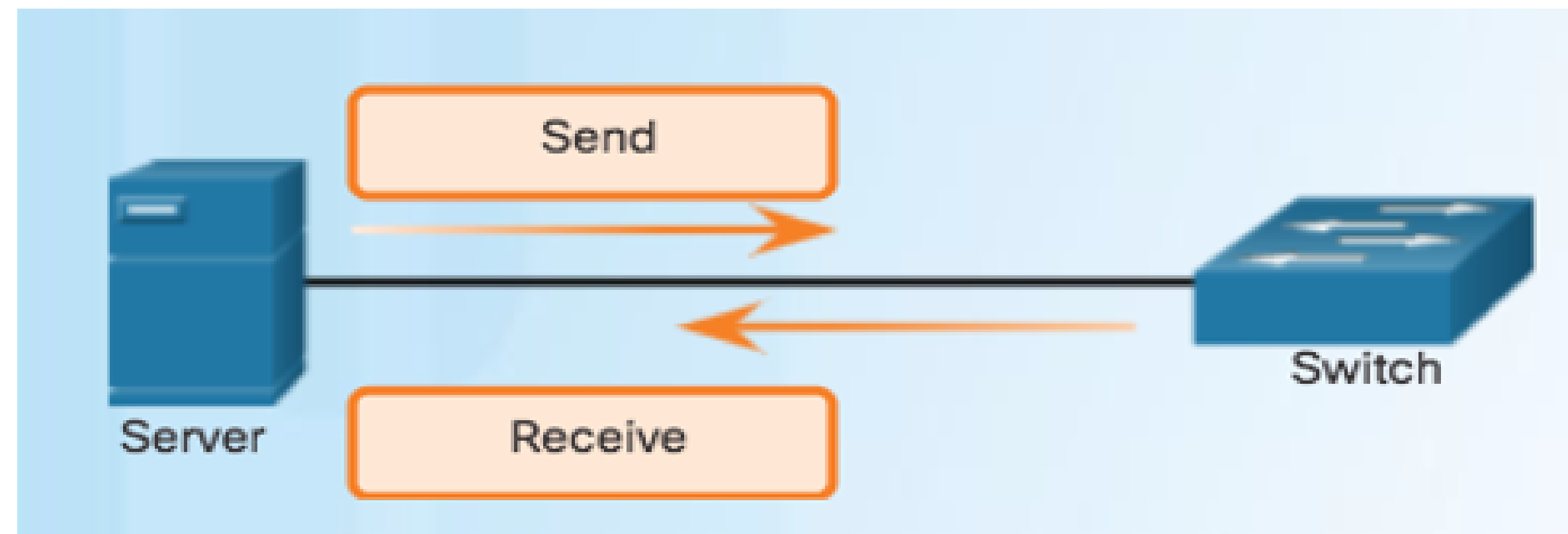
■ Half-Duplex Communication

- Both devices can transmit and receive on the media but cannot do so simultaneously.
- Used in legacy bus topologies and with Ethernet hubs.
- WLANs also operate in half-duplex.

LAN – Communication

LAN Topologies

Half and Full Duplex (Cont.)



■ Full-Duplex Communication

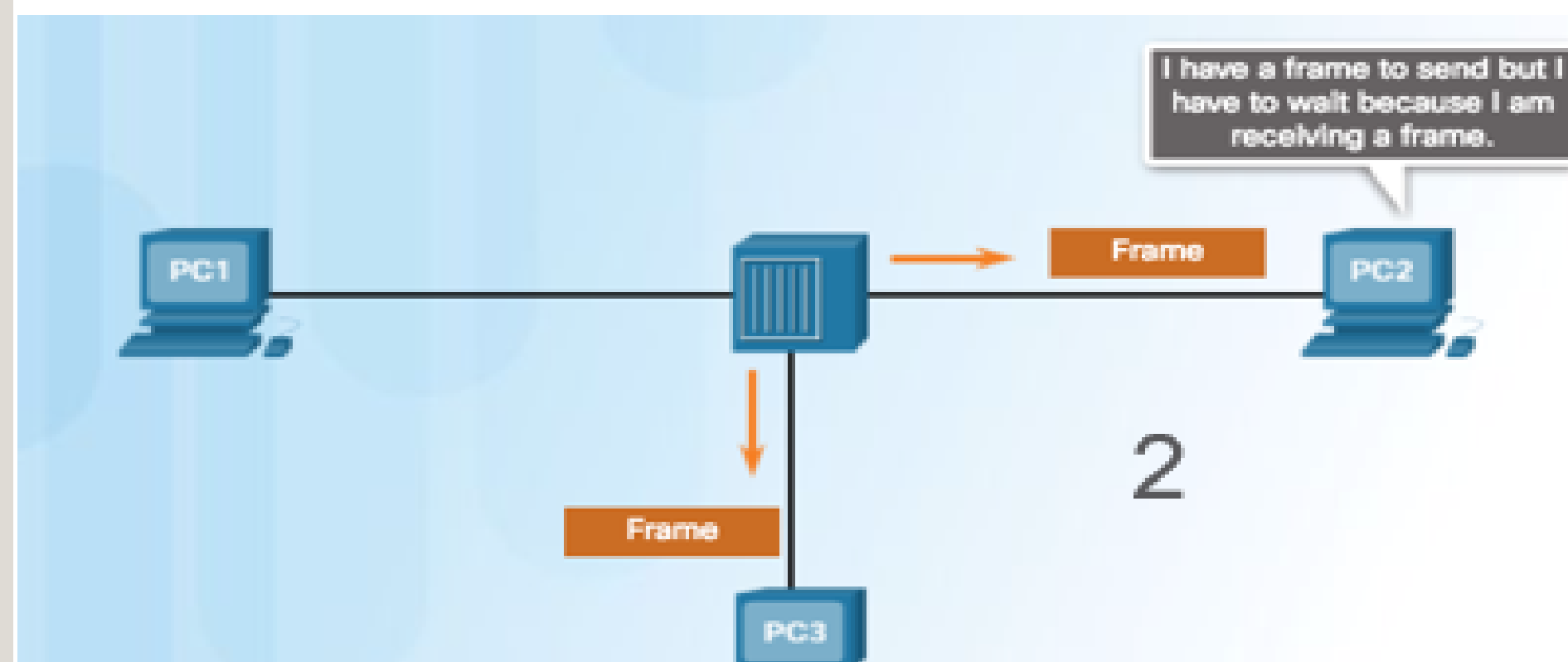
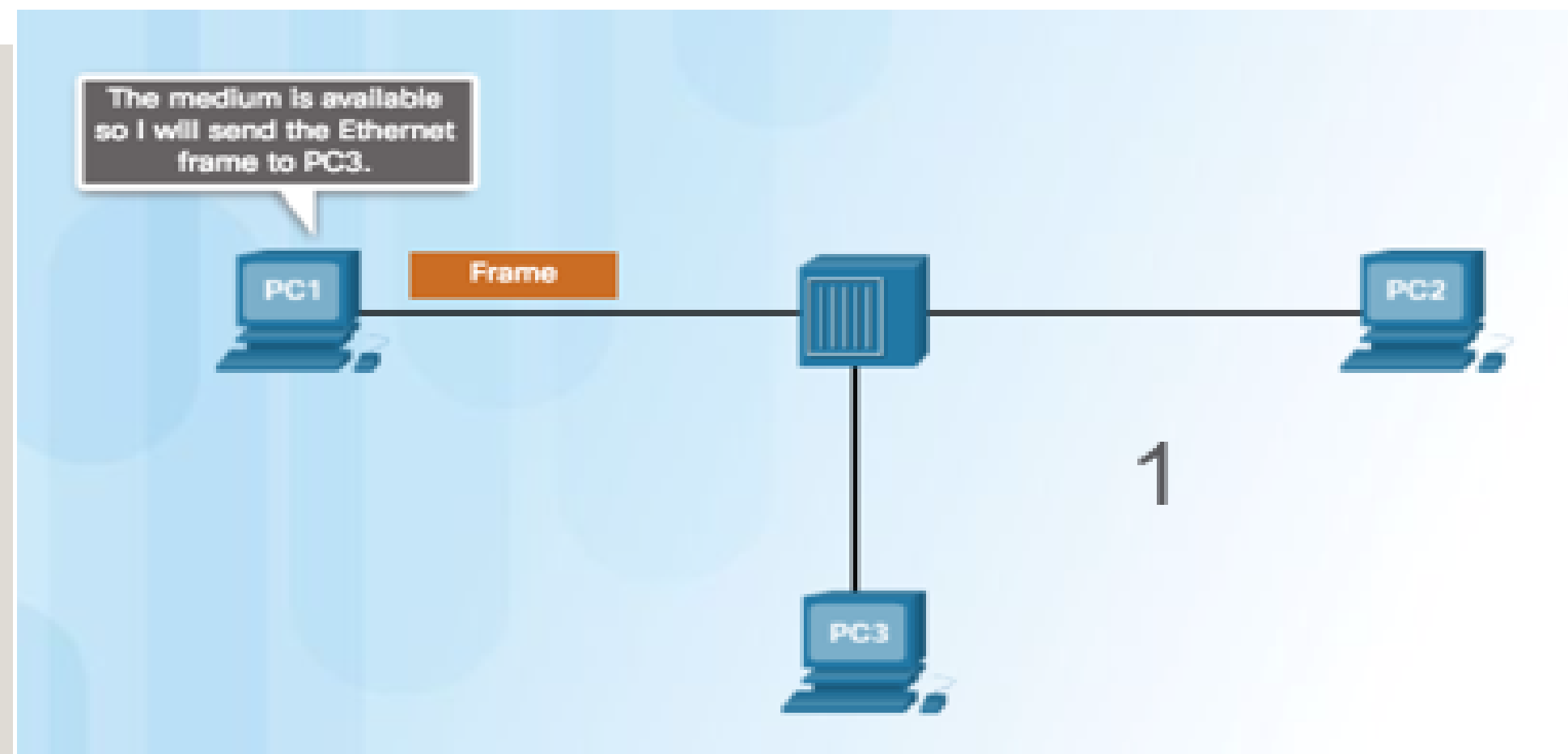
- Both devices can transmit and receive on the media at the same time.
- Data link layer assumes that the media is available for transmission for both nodes at any time.
- Ethernet switches operate in full-duplex mode by default, but can operate in half-duplex if connecting to a device such as an Ethernet hub.

LAN – CSMA-CD

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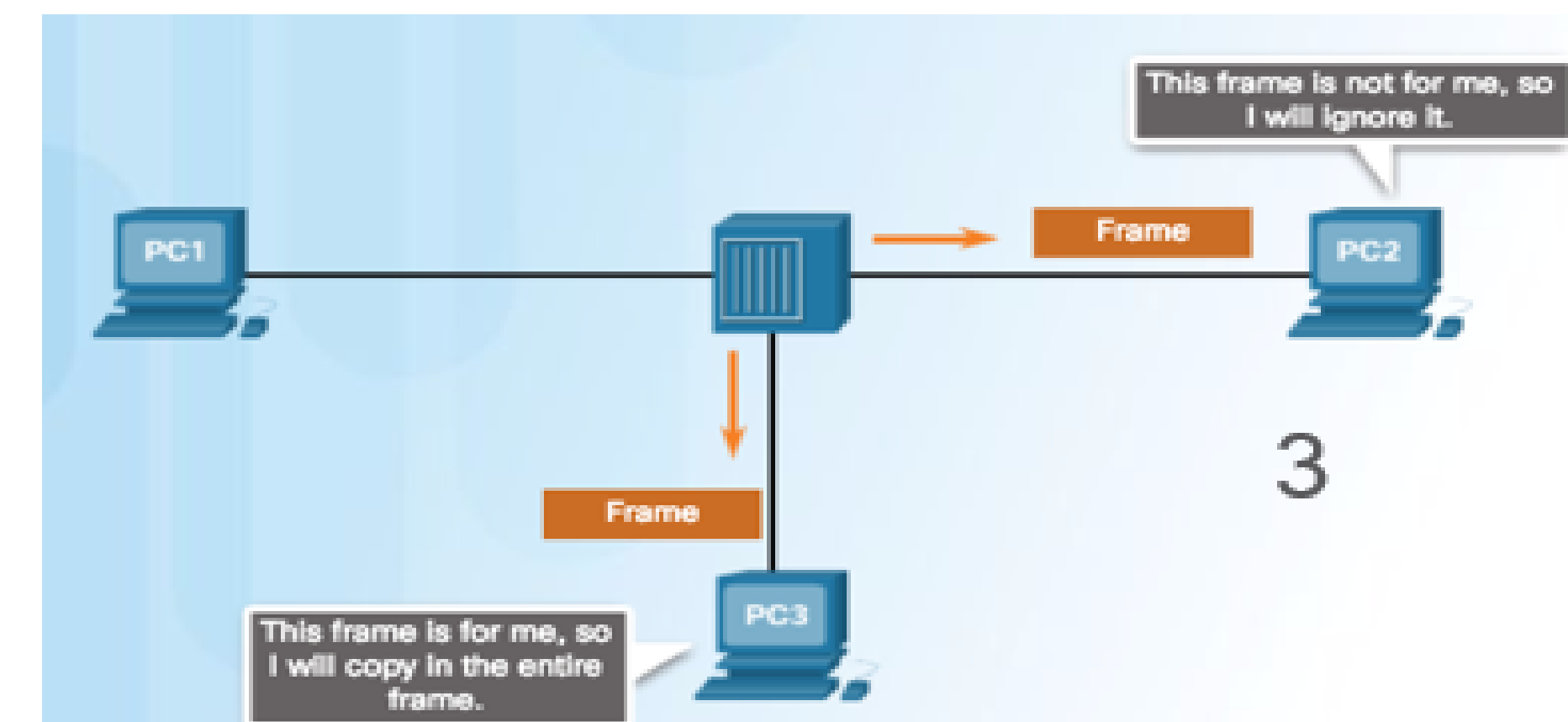
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Contention-based Access - CSMA/CD



- Carrier Sense Multiple Access/Collision Detection (CSMA/CD) process is used in half-duplex Ethernet LANs.

- If two devices transmit at the same time, a collision will occur.
- Both devices will detect the collision on the network.
- Data sent by both devices will be corrupted and will need to be resent.

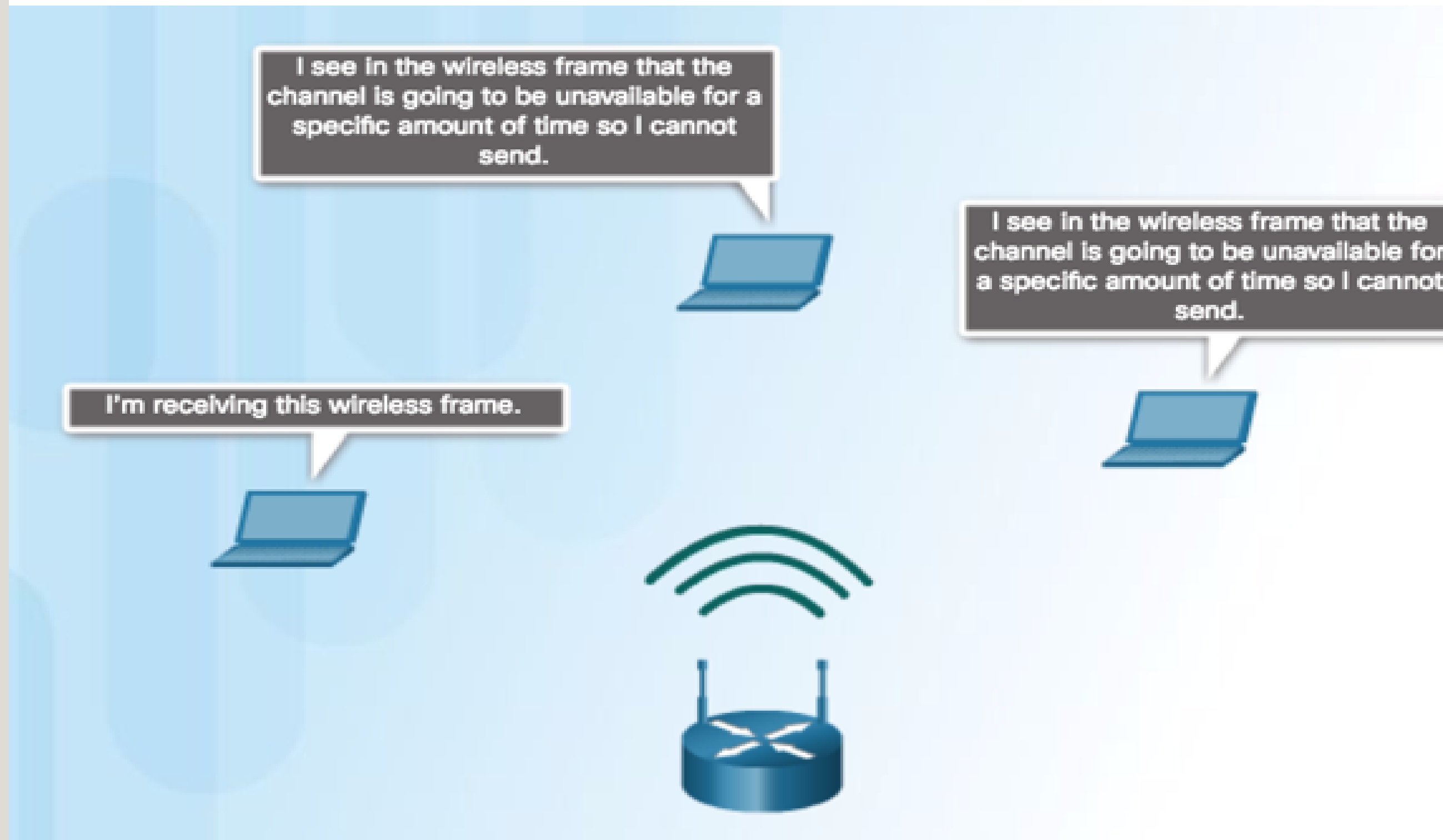


LAN – CSMA CA

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Contention-based Access - CSMA/CA



▪ CSMA/CA

- Uses a method to detect if the media is clear.
- Does not detect collisions but attempts to avoid them by waiting before transmitting.
- **Note:** Ethernet LANs using switches do not use a contention-based system because the switch and the host NIC operate in full-duplex mode.

Reference Model

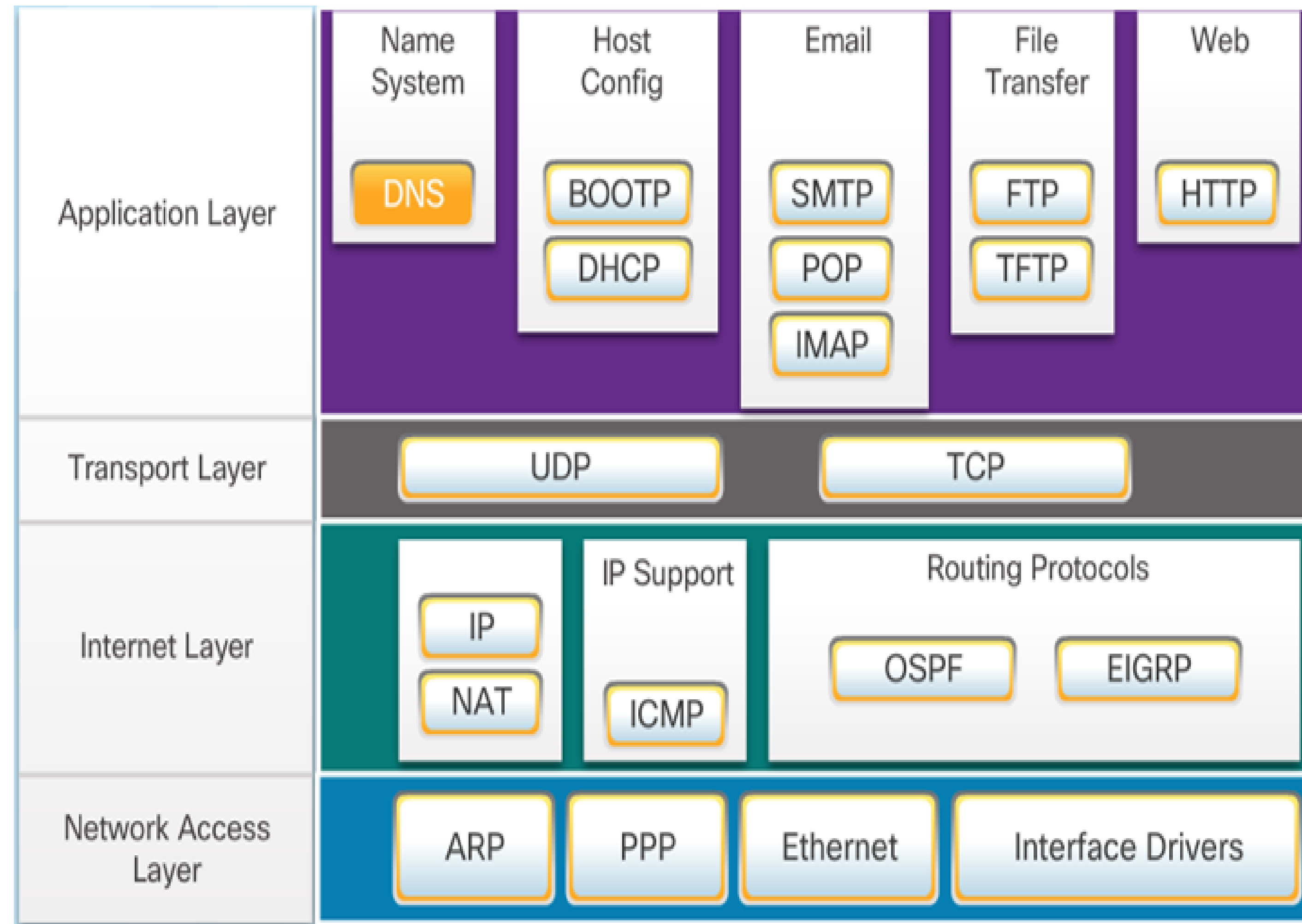
- **The first**, and the more theoretically general, is ISO/**OSI** (International Organization for Standardization/Open Systems Interconnection).
- **The second**, and the more practical, is Transmission Control Protocol/Internet Protocol (**TCP/IP**) which **minimizes** the number of layers used and makes more focus on the practical issues than the theoretical issues.

REFERENCE MODEL

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TCP/IP Protocol Suite

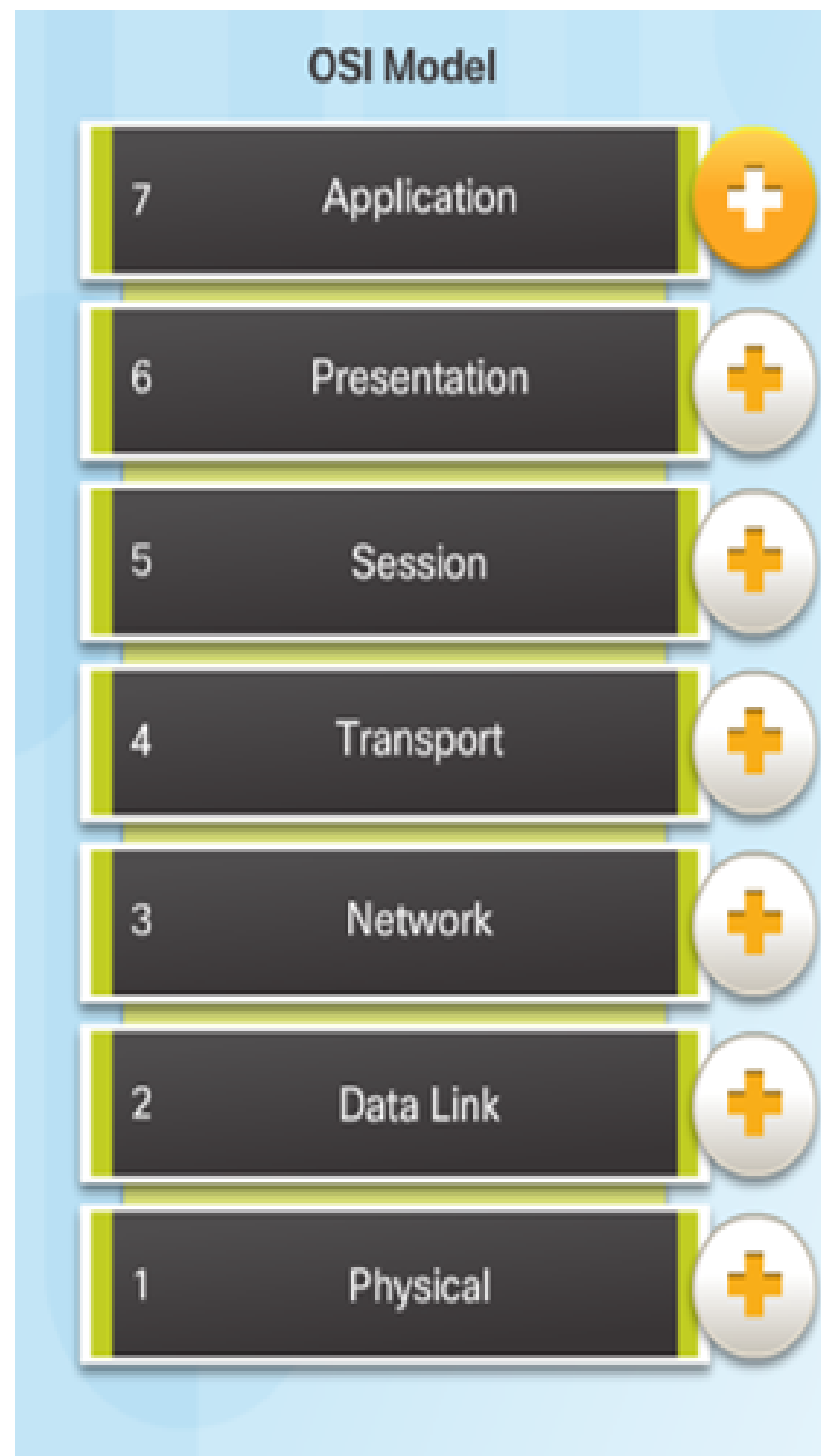


REFERENCE MODEL

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The OSI Reference Model



- Application - contains protocols used for process-to-process communications.
- Presentation - provides for common representation of the data.
- Session - provides services to the presentation layer to organize its dialogue and to manage data exchange.
- Transport - defines services to segment, transfer, and reassemble the data.
- Network - provides services to exchange the individual pieces of data over the network between identified end devices.
- Data Link - provides methods for exchanging data frames between devices over a common media.
- Physical - describes the mechanical, electrical, functional, and procedural means to transmit bits across physical connections.

REFERENCE MODEL

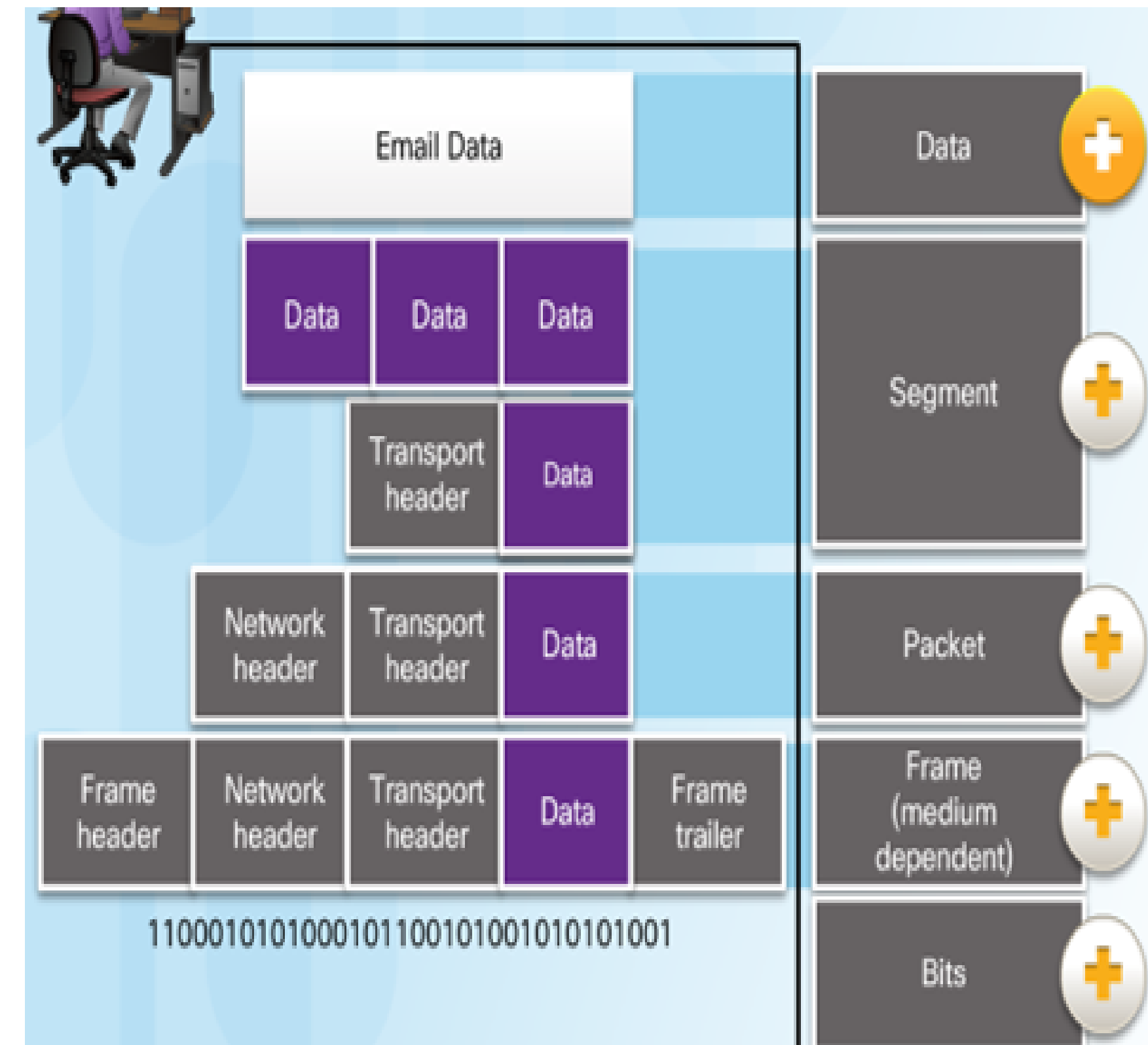
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Data Encapsulation

Protocol Data Units

- As application data is passed down the protocol stack, information is added at each level. This is known as the **encapsulation** process.
- The form that the data takes at each layer is known as a Protocol Data Unit (PDU).
 - Data - application layer PDU
 - Segment – Transport layer PDU
 - Packet – Network layer PDU
 - Frame – Data Link Layer PDU
 - Bits – Physical Layer PDU



REFERENCE MODEL

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Data Encapsulation

Encapsulation Example

- The encapsulation process works from top to bottom:
 - Data is divided into segments.
 - The TCP segment is encapsulated in the IP Packet.
 - The IP packet is encapsulated in the Ethernet Frame.

